

解析函数零点的孤立性 (证明).

解析函数唯一性定理

解析函数无零因子.

洛必达法则.

$$\text{e.g. } \lim_{z \rightarrow 0} \frac{\sin z^n}{z^m} = \begin{cases} 0 & n > m \\ 1 & n = m \\ \infty & n < m \end{cases}$$

孤立奇点的分类: 可去奇点 / 极点 / 本性奇点
(L-级数).

例 1. $f(z) = \frac{\sin z}{z^n}$

当 $n=1$ 时, $f(z) = \frac{\sin z}{z} = \sum_{n=1}^{+\infty} \frac{(-1)^{n-1} z^{2n-1}}{(2n-1)! \cdot z}$ (泰勒)

$$= \sum_{n=1}^{+\infty} \frac{(-1)^{n-1} z^{2(n-1)}}{(2n-1)!} = \sum_{m=0}^{+\infty} \frac{(-1)^m z^{2m}}{(2m+1)!}$$

收敛半径为无穷大.

当 $n \neq 1$ 时, $f(z) = \frac{\sin z}{z^n}, n \geq 2$.

 $z=0$ 时 $n-1$ 阶极点.

$$f(z) = \frac{1}{z^{n-1}} + \sum_{k=2}^{+\infty} \frac{(-1)^k z^{2k-n-1}}{(2k-1)!}.$$

(省略一些奇点的例子).

Picard 大定理.

如果 $f(z)$ 有一个本性奇点 (z_0) , 令 v 为 z_0 的一个邻域.

对任意的 $w \in \mathbb{C}$, 除了有一个 w 可能无法取到 $f(z)=w$, 其余的 w 都有解
且有无穷多解.

Picard 小定理.

若 $f(z)$ 处处可导, 且不是常数, 则至多有一个 $w \in \mathbb{C}$, 使得 $\forall z \in \mathbb{C}, f(z) \neq w$.

Liouville 定理.

有界的整函数是常数.

例 2. $\oint_{|z|=1} \frac{\sin z}{z} dz$.

$$\oint_{|z|=1} \frac{\sin z}{z} dz = 2\pi i \cdot C_{-1}.$$

而 $\frac{\sin z}{z} = \sum_{m=0}^{+\infty} \frac{(-1)^m z^{2m}}{(2m+1)!}$ $C_{-n}=0$ 故 $C_{-1}=0$. $\oint_{|z|=1} \frac{\sin z}{z} dz = 0$.

例3. 求积分 $I_n = \oint_{|z|=R>r} \frac{1}{r^n + z^n} dz$. $r>0$. $n \in \mathbb{N}^*$.
(重点)

另:
 $\text{Res}[f, z_k]$
 $= C_{-1}^{(k)}$

$$I_n = 2\pi i \left(\sum_{k=1}^n \text{Res} \left[\frac{1}{r^n + z^n}, z_k \right] \right).$$

$$= 2\pi i \left(\sum_{k=1}^n \frac{1}{n z_k^{n-1}} \right).$$

其中, z_k 满足 $z_k^n + r^n = 0 \Rightarrow -z_k = \frac{r^n}{z_k^{n-1}} \Rightarrow \frac{1}{z_k^{n-1}} = \frac{-z_k}{r^n}$

故 $I_n = 2\pi i \left(\sum_{k=1}^n \frac{-z_k}{r^n} \right) = (2\pi i) \frac{(-1)}{r^n} \left(\sum_{k=1}^n z_k \right).$

当 $n=1$ 时, $I_n = \oint_{|z|=R>r} \frac{1}{z+r} dz = \oint_{|z|=R>r} \frac{f(z)}{z-(r)}$ $= 2\pi i$.

当 $n>2$ 时, $z^n + r^n = (z-z_1)(z-z_2)\dots(z-z_n)$

$$= z^n - \left(\sum_{k=1}^n z_k \right) z^{n-1} + \dots + (-1)^n \prod_{k=1}^n z_k.$$

若 $n>2$, $\sum_{k=1}^n z_k = 0$, 那么 $I_n = 0$

综上, $I_n = \begin{cases} 0 & n>2 \\ 2\pi i & n=1 \end{cases}$

(省略柯西-xxx定理暴力求解法)

例4. 求积分 $I_n = \oint_{|z|=1} \frac{1 - \cos 4z^3}{z^n} dz$. $n \in \mathbb{N}^*$.

$$I_n = \oint_{|z|=1} \frac{1 - \sum_{m=0}^{+\infty} \frac{(-1)^m (4z^3)^{2m}}{(2m)!}}{z^n} dz.$$

$$= \sum_{m=1}^{+\infty} \frac{(-1)^{m+1} 4^{2m}}{(2m)!} \oint_{|z|=1} z^{6m-n} dz.$$

对于 $\oint_{|z|=1} z^{6m-n} dz$, 有原式 = $\begin{cases} 0 & n \neq 6m+1, m=1,2,\dots \\ 2\pi i & n=6m+1, m=1,2,\dots \end{cases}$

$\hookrightarrow$ 即 $m = \frac{n-1}{6}$.

代入, 有 $I_n = \begin{cases} 0 & n \neq 6m+1, m=1,2,\dots \\ (-1)^{\frac{n+1}{6} + \frac{n-1}{3}} \frac{2\pi i}{(\frac{n-1}{3})!} & n=6m+1 \end{cases}$

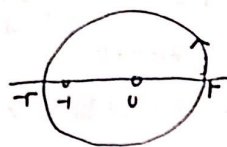
变体=例如

$$\oint_{|z|=1} \frac{1 - \cos az^b}{z^n} dz.$$

课上 $a=4, b=3$ 的情况

例 15. 求积分 $I = \oint_{|z|=r>1} \frac{z^3 e^{\frac{1}{z}}}{1+z} dz$

法 1 = (留数).



$$\text{原式} = 2\pi i \left(\text{Res} \left[\frac{z^3 e^{\frac{1}{z}}}{1+z}, 0 \right] + \text{Res} \left[\frac{z^3 e^{\frac{1}{z}}}{1+z}, -1 \right] \right).$$

$$\text{其中: } \text{Res} \left[\frac{z^3 e^{\frac{1}{z}}}{1+z}, -1 \right] = \frac{(-1)^3 e^{-1}}{(1+z)'|_{z=-1}} = (-1)^3 e^{-1} = -e^{-1} = -\frac{1}{e}.$$

$$\text{而又 } \frac{z^3 e^{\frac{1}{z}}}{1+z} = \sum_{k=-\infty}^{+\infty} c_k z^k \quad \text{求 } c_{-1} = ?.$$

$$\frac{z^3 e^{\frac{1}{z}}}{z+1} = z^3 \left(1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \dots \right) (1 - z + z^2 - z^3 + z^4 - \dots).$$

即求 $(1 + \frac{1}{z} + \frac{1}{2!z^2} + \dots)(1 - z + z^2 - z^3 + \dots)$ 中 z^{-4} 的系数.

$$c_{-1} = \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!} + \dots \quad (\text{写成这个式子也算对, 但还可化简}).$$

$$\text{由于 } e^{-1} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \dots = \frac{1}{2!} - \frac{1}{3!} + c_{-1}.$$

$$\text{故 } c_{-1} = e^{-1} - \frac{1}{3}.$$

$$\text{综上, 原式} = 2\pi i (e^{-1} - \frac{1}{3} - \frac{1}{e}) = -\frac{2}{3}\pi i.$$

(法 2 是用柯西高阶导数公式算 $\text{Res} \left[\frac{z^3 e^{\frac{1}{z}}}{z-(-1)}, -1 \right]$ (忽略)) $\times$.

法 3 是利用倒数:

$$\text{又对于 } I = \oint_{|z|=r>1} \frac{z^3 e^{\frac{1}{z}}}{1+z} dz.$$

$$\text{令 } \frac{1}{z} = t \Rightarrow dz = d\left(\frac{1}{t}\right) = -\frac{dt}{t^2}, \quad |z|=r < \Rightarrow |t| = \frac{1}{r} < 1$$

由于积分方向变换.

$$I = - \oint_{|t|=\frac{1}{r}<1} \frac{e^t}{t^3(1+t)} \cdot (-\frac{1}{t^2}) dt.$$

$$= \oint_{|t|=\frac{1}{r}<1} \frac{e^t}{t^4(1+t)} = 2\pi i \text{Res} \left[\frac{e^t}{t^4(1+t)}, 0 \right].$$

$$\text{而 } \frac{e^t}{t^4(1+t)} \text{ 求 } t^{-1} \text{ 的系数 } c_{-1}. \text{ 即求 } \frac{e^t}{1+t} \text{ 的 } t^3 \text{ 的系数 } t^3.$$

$$\text{而 } \frac{e^t}{1+t} = \left(1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots \right) (1 - t + t^2 - t^3 + \dots).$$

$$t^3 \text{ 的系数为: } \frac{1}{3!} - \frac{1}{2!} + 1 - 1 = -\frac{1}{3}.$$

$$\text{故 } c_{-1} = -\frac{1}{3}.$$

$$\text{故 } I = -\frac{1}{3} 2\pi i = -\frac{2}{3}\pi i.$$

例6. $I_n = \oint_{|z|=R>r} \frac{1}{r^n + z^n} dz \quad n \in \mathbb{N}, r > 0$

(法一: 见例三留数法)

法二: 令 $z = \frac{1}{t}$, $dz = (-\frac{1}{t^2})dt$.

$|z|=R \Leftrightarrow |t| = \frac{1}{R} < \frac{1}{r}$

那么, $I_n = - \oint_{|t|=\frac{1}{R} < \frac{1}{r}} (-\frac{1}{t^2}) \frac{1}{r^n + \frac{1}{t^n}} dt = \oint_{|t|=\frac{1}{R} < \frac{1}{r}} \frac{t^{n-2}}{(t^n r^n + 1)} dt$

由于 $t^n r^n + 1 \neq 0$.

$n=1$ 时, $I_n = 2\pi i$.

$n \geq 2$ 时, $\frac{t^{n-2}}{t^n r^n + 1}$ 没有奇点, $I_n = 0$.

综上, $I_n = \begin{cases} 2\pi i & n=1 \\ 0 & n \geq 2 \end{cases}$

(下面计算实积分)

例7. 求 $I_{a,b} = \int_0^{2\pi} \frac{d\theta}{a + b \cos \theta}$, 其中 $a > |b| > 0$, $a, b \in \mathbb{R}$. [三角积分]

例7. $I_{a,b} = \int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = \int_0^{2\pi} \frac{d\theta}{a + b \sin \theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}$.

结论: 更一般地, 若 f 在 $[-1, 1]$ 上连续可积, 那么 $\int_0^{2\pi} f(\cos \theta) d\theta = \int_0^{2\pi} f(\sin \theta) d\theta$.

当 $b=0$ 时, $I_{a,b} = \int_0^{2\pi} \frac{1}{a} d\theta = \frac{2\pi}{a}$.

当 $b \neq 0$ 时, 由于 $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$, $\sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$,
再令 $z = e^{i\theta}$, $\Rightarrow |z|=1$, $dz = ie^{i\theta} d\theta = iz d\theta$.

因此, 转化为复积分:

$d\theta = \frac{dz}{iz}$, $\cos \theta = \frac{1}{2}(z + z^{-1}) = \frac{z^2 + 1}{2z}$. $\sin \theta = \frac{z^2 - 1}{2iz}$.

所以, $I_{a,b} = \oint_{|z|=1} \frac{1}{(a + b(\frac{z^2 + 1}{2z}))iz} dz$.

$= \frac{2}{i} \oint_{|z|=1} \frac{dz}{bz^2 + 2az + b}$

由于 $b \neq 0$, $I_{a,b} = \frac{2}{ib} \oint_{|z|=1} \frac{dz}{z^2 + 2\frac{a}{b}z + 1}$

考虑 $z^2 + 2\frac{a}{b}z + 1 = 0$, 有 $z_{1,2} = -\frac{a}{b} \pm \sqrt{\frac{a^2}{b^2} - 1} = \frac{-a \pm \sqrt{a^2 - b^2}}{b}$

不妨设 $b > 0$,

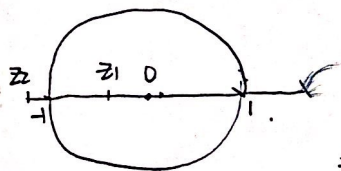
$z_2 = \frac{-a - \sqrt{a^2 - b^2}}{b} < -\frac{a}{b} < -1$. $z_1 z_2 = 1$.

故 $-1 < z_1 < 0$.

原被积函数在 $|z|=1$ 圆盘内仅有 z_1 一个奇点,

因此, $I_{a,b} = \frac{2}{ib} \oint_{|z|=1} \frac{1}{(z^2 + 2\frac{a}{b}z + 1)} \cdot 2\pi i$

$= \frac{2}{ib} \cdot \frac{1}{z_1 + \frac{a}{b}} \pi i$



$$I_{a,b} = \frac{2\pi}{b(z_1 + \frac{a}{b})} = \frac{2\pi}{b\sqrt{a^2 - b^2}} = \frac{2\pi}{\sqrt{a^2 - b^2}}$$

例8. 求 $I_p = \int_0^{2\pi} \frac{d\theta}{1 - 2p\cos\theta + p^2}$, $p \in (-1, 1)$

$$(I_p = \int_0^{2\pi} \frac{d\theta}{1 - 2p\cos\theta + p^2} = \int_0^{2\pi} \frac{d\theta}{1 - 2p\sin\theta + p^2})$$

$$I_p = \int_0^{2\pi} \frac{d\theta}{(p^2+1) - 2p\cos\theta}, p \in (-1, 1).$$

$$(p^2+1 > 2p, p \in (-1, 1))$$

$$= \frac{2\pi}{\sqrt{(p^2+1)^2 - (2p)^2}} = \frac{2\pi}{1-p^2}.$$

(利用例7的结论. 先证 $\frac{2\pi}{\sqrt{a^2-b^2}}$, 再应用).

例9. $I_{A,B} = \int_0^{2\pi} \frac{d\theta}{A^2\cos^2\theta + B^2\sin^2\theta}$, $A, B > 0$.

$$\text{由于令 } \cos^2\theta = \frac{1 + \cos 2\theta}{2}$$

$$\sin^2\theta = \frac{1 - \cos 2\theta}{2}$$

$$\text{那么, } I_{A,B} = \int_0^{2\pi} \frac{2}{A^2(1 + \cos 2\theta) + B^2(1 - \cos 2\theta)} d\theta.$$

$$= \int_0^{2\pi} \frac{2}{(A^2+B^2) + (A^2-B^2)\cos 2\theta} d\theta.$$

$$\text{令 } \varphi = 2\theta, d\varphi = 2d\theta, d\theta = \frac{1}{2}d\varphi.$$

$$I_{A,B} = \int_0^{4\pi} \frac{d\varphi}{(A^2+B^2) + (A^2-B^2)\cos\varphi}$$

$$= 2 \int_0^{2\pi} \frac{d\varphi}{(A^2+B^2) + (A^2-B^2)\cos\varphi}$$

$$\text{令 } a = A^2+B^2, b = A^2-B^2, a > |b| > 0.$$

$$I_{A,B} = 2 \frac{2\pi}{\sqrt{a^2-b^2}} = \dots = \frac{2\pi}{AB}$$

这里代进去算

(先证例7. 再应用).

例10. 计算 $J_{a,b} = \int_0^{2\pi} \frac{d\theta}{(a+b\cos\theta)^2}$ (似乎没在课上讲此题)

$$\text{已有 } I_{a,b} = \int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{\sqrt{a^2-b^2}} = 2\pi(a^2-b^2)^{-1/2}.$$

将 a 当作参数, 将 b 当作常数. 两边同时对 a 求导:

$$\frac{d}{da}(I_{a,b}) = \int_0^{2\pi} \frac{(-1)d\theta}{(a+b\cos\theta)^2}$$

$$\text{故 } J_{a,b} = -2\pi(-\frac{1}{2})(2a)(a^2-b^2)^{-3/2} = \frac{2\pi a}{(a^2-b^2)^{3/2}}$$

$$\text{更一般地, } J_{(a)}^{(n)} = \int_0^{2\pi} \frac{d\theta}{(a+b\cos\theta)^n}$$

两边对 a 求导, $\frac{dJ^{(n)}(a)}{da} = \int_0^{2\pi} \frac{(-n) d\theta}{(a+b\cos\theta)^{n+1}}$

得到递推公式: $J^{(n+1)}(a) = -\frac{1}{n} \frac{dJ^{(n)}(a)}{da}$

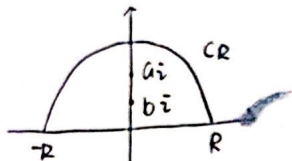
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0 $f(z) = \frac{1}{1+z^2} = \sum_{n=0}^{\infty} (-1)^n z^{2n}$

例 11. 计算 $\int_0^{+\infty} \frac{1}{(x^2+a^2)(x^2+b^2)} dx, a>0, b>0$

$I = \int_0^{+\infty} \frac{1}{(x^2+a^2)(x^2+b^2)} dx = \frac{1}{2} \lim_{R \rightarrow +\infty} \int_{-R}^R \frac{dx}{(x^2+a^2)(x^2+b^2)}$

令 $R > \max\{a, b\}$.



有 $\int_{-R}^R \frac{1}{(x^2+a^2)(x^2+b^2)} dx + \int_{CR} \frac{1}{(z^2+a^2)(z^2+b^2)} dz = 2\pi i (C_{-1}^{(1)} + C_{-1}^{(2)})$

其中, $\int_{CR} \frac{1}{(z^2+a^2)(z^2+b^2)} dz$ 趋于零, $C_{-1}^{(1)}, C_{-1}^{(2)}$ 分别对应 ai, bi 处的留数

$C_{-1}^{(1)} = \text{Res} \left(\frac{1}{(z^2+a^2)(z^2+b^2)}, ai \right)$
 $= \text{Res} \left[\frac{1}{z^2 + (a^2+b^2)z^2 + a^2b^2}, ai \right]$

$= \frac{1}{4z^3 + z(a^2+b^2)z} \Big|_{z=ai} = \frac{1}{4(ai)^3 + z(a^2+b^2)(ai)}$
 $= \frac{1}{2ai(b^2-a^2)}$

交换 a, b 位置, 有:

$C_{-1}^{(2)} = \frac{1}{2bi(a^2-b^2)}$

故 $\int_0^{+\infty} \frac{1}{(x^2+a^2)(x^2+b^2)} dx = \frac{1}{2} \int_{-R}^R \frac{1}{(x^2+a^2)(x^2+b^2)} dx = \pi i (C_{-1}^{(1)} + C_{-1}^{(2)})$

$= \pi i \left(\frac{1}{2ai(b^2-a^2)} + \frac{1}{2bi(a^2-b^2)} \right)$

$= \frac{\pi}{2} \left[\frac{b-a}{(b^2-a^2)ab} \right] = \frac{\pi}{2} \left(\frac{1}{b} - \frac{1}{a} \right)$

$= \frac{\pi}{2ab(a+b)}$

(当 $a=b$ 时, $\int_0^{+\infty} \frac{dx}{(x^2+a^2)^2} = \frac{\pi}{4a^3}$)

变形: $\int_0^{+\infty} \frac{x^2 dx}{(x^2+a^2)(x^2+b^2)}, a>0, b>0$

例12. 求 $\int_0^{+\infty} \frac{x \sin kx}{(x^2+a^2)(x^2+b^2)} dx, a>0, b>0, k>0$

$$I = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{x \sin kx}{(x^2+a^2)(x^2+b^2)} dx = \frac{1}{2i} \lim_{R \rightarrow +\infty} \int_{-R}^R \frac{x e^{ikx}}{(x^2+a^2)(x^2+b^2)} dx$$

(这是因为 $\int_{-\infty}^{+\infty} \frac{x \cos kx}{(x^2+a^2)(x^2+b^2)} dx$ 是奇函数)

$$I = \frac{1}{2i} \int_{-\infty}^{+\infty} \frac{x e^{ikx}}{(x^2+a^2)(x^2+b^2)} dx = \frac{2\pi i}{2i} \left(\text{Res} \left[\frac{z e^{ikz}}{z^4+(a^2+b^2)z^2+a^2b^2}, ai \right] + \text{Res} \left[\frac{z e^{ikz}}{z^4+(a^2+b^2)z^2+a^2b^2}, bi \right] \right)$$

$$= \pi \left(\frac{z e^{ikz}}{4z^3+(a^2+b^2)(2z)} \Big|_{z=ai} + \frac{z e^{ikz}}{4z^3+(a^2+b^2)(2z)} \Big|_{z=bi} \right)$$

$$= \pi \left(\frac{a i e^{-ka}}{4(a i)^3+(a^2+b^2)(2a i)} + \frac{b i e^{-kb}}{4(b i)^3+(a^2+b^2)(2b i)} \right)$$

$$= \frac{\pi}{2(b^2-a^2)} (e^{-ka} - e^{-kb})$$

如果 $b \rightarrow a$ 时, 把 a 看作常数, 对 b 使用洛必达法则

$$\int_0^{+\infty} \frac{x \sin kx}{(x^2+a^2)^2} dx = \frac{\pi k e^{-ka}}{4a}$$

例13. 求 $\int_0^{+\infty} \frac{x^3 \sin kx}{(x^2+a^2)(x^2+b^2)} dx, k>0$

与例12完全相同, 除求留数处

$$\text{结论: } I = \frac{b^2 e^{-kb} - a^2 e^{-ka}}{2(b^2-a^2)} \pi$$

$$\text{当 } b \rightarrow a \text{ 时, } \int_0^{+\infty} \frac{x^3 \sin kx}{(x^2+a^2)^2} dx = \frac{(2-ak)e^{-ka}}{4} \pi$$

$$\text{当 } a \rightarrow 0^+ \text{ 时, } \int_0^{+\infty} \frac{\sin kx}{x} dx = \begin{cases} \frac{\pi}{2} & k>0 \\ 0 & k=0 \\ -\frac{\pi}{2} & k<0 \end{cases}$$

$$\Delta \int_0^{+\infty} \frac{\sin kx}{x} dx$$

$$\Delta \text{ 变形: } \int_0^{+\infty} \frac{x^2 \cos kx}{x^2+a^2(x^2+b^2)} dx$$

参考例18.

$$\int_0^{+\infty} \frac{\sin kx}{x} dx$$

例14. 求 $\int_0^{+\infty} \frac{x \sin kx}{x^2+a^2} dx$

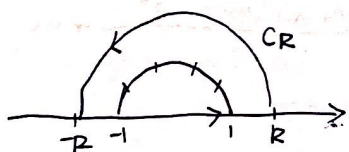
与例12完全相同, 除只求一个留数

$$\text{结论: } I = \frac{\pi}{2e^{ka}} = \frac{\pi}{2} e^{-ka}$$

$$\text{当 } a \rightarrow 0^+ \text{ 时, } \int_0^{+\infty} \frac{\sin kx}{x} dx = \begin{cases} \frac{\pi}{2} & k>0 \\ 0 & k=0 \\ -\frac{\pi}{2} & k<0 \end{cases}$$

例15. 求 $I_n = \int_0^{+\infty} \frac{1}{1+x^{2n}} dx, n=1,2,\dots$

$$\text{结论: } I_n = \frac{\pi}{2n \sin \frac{\pi}{2n}}$$



(图也算分)

$$\int_R \frac{1}{1+z^{2n}} dz + \int_{c_R} \frac{1}{1+z^{2n}} dz = 2\pi i \sum_{k=1}^n \text{Res} \left[\frac{1}{1+z^{2n}}, z_k \right]$$

$$= 2\pi i \left(\sum_{k=1}^n \frac{1}{2n z_k^{2n-1}} \right) = \frac{\pi i}{n} \left(\sum_{k=1}^n \frac{1}{z_k^{2n-1}} \right)$$

其中 $\int_{c_R} \frac{1}{1+z^{2n}} dz$ 趋于0

$$\text{又由于 } z_k^n = -1, -z_k = \frac{1}{z_k^{2n-1}}$$

$$\text{故原式} = -\frac{\pi i}{n} \left(\sum_{k=1}^n z_k \right)$$

$$\text{由 } z_k^n = -1 = e^{\pi i} = e^{\pi i + 2(k-1)\pi i} = e^{(2k-1)\pi i}$$

$$\text{故 } z_k = e^{\frac{(2k-1)\pi i}{2n}} = e^{\frac{k\pi i}{n}} / e^{\frac{\pi i}{2n}} \quad k=1, 2, \dots, n.$$

$$\text{因此, } \sum_{k=1}^n z_k = \frac{\sum_{k=1}^n e^{\frac{k\pi i}{n}}}{e^{\frac{\pi i}{2n}}} = \frac{e^{\frac{\pi i}{n}} (1 - e^{\pi i})}{e^{\frac{\pi i}{2n}} (1 - e^{\frac{\pi i}{n}})} = \frac{2e^{\frac{\pi i}{2n}}}{1 - e^{\frac{\pi i}{n}}}$$

$$= \frac{2e^{\frac{i\theta}{2}}}{1 - e^{i\theta}} \quad (\text{其中 } \theta = \frac{\pi}{2n})$$

$$= \frac{2(\cos\theta + i\sin\theta)}{1 - \cos\theta - i\sin\theta}$$

$$= \frac{2(\cos\theta + i\sin\theta)}{2\sin^2\theta - 2i\sin\theta\cos\theta} = \frac{\cos\theta + i\sin\theta}{\sin\theta(\sin\theta - i\cos\theta)}$$

$$= \frac{1}{(-i)\sin\theta} = \frac{1}{-i\sin\frac{\pi}{2n}} = \frac{i}{\sin\frac{\pi}{2n}}$$

$$\text{故原式} = -\frac{\pi i}{n} \cdot \frac{i}{\sin\frac{\pi}{2n}} = \frac{\frac{\pi}{n}}{\sin\frac{\pi}{2n}}$$

$$I_n = \frac{1}{2} \int_0^{+\infty} \frac{dx}{1+x^{2n}} = \frac{\frac{\pi}{2n}}{\sin\frac{\pi}{2n}}.$$

$$\text{更一般地, } I_{1/n} = \int_0^{+\infty} \frac{dx}{1+x^{2n}} = \frac{1}{1^{2n-1}} I_n = \frac{1}{1^{2n-1}} \cdot \frac{\pi}{2n} \cdot \frac{1}{\sin\frac{\pi}{2n}}$$

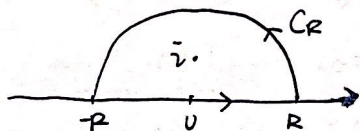
$$I_1 = \frac{\pi}{2}, I_2 = \frac{\pi}{2\sqrt{2}}, I_3 = \frac{\pi}{8}, \dots, I_1 > I_2 > I_3 > \dots > \frac{0^+}{\sin 0^+} = 1.$$

$$\text{例 11b. 求 } J_n = \int_0^{+\infty} \frac{dx}{(1+x^2)^n}, \quad n=1, 2, \dots$$

$$\text{令 } R > 1.$$

$$\int_{-R}^R \frac{1}{(1+x^2)^n} dx + \int_{C_R} \frac{dz}{(1+z^2)^n} = 2\pi i \operatorname{Res}\left[\frac{1}{(1+z^2)^n}, i\right]$$

$$\text{令 } R \rightarrow +\infty, \int_{C_R} \frac{dz}{(1+z^2)^n} \rightarrow 0.$$



$$\text{故 } J_n = \frac{1}{2} \int_0^{+\infty} \frac{1}{(1+x^2)^n} dx = \frac{1}{2} \int_{C_R} \frac{1}{(1+z^2)^n} dz = \frac{1}{2} \lim_{R \rightarrow +\infty} \int_{-R}^R \frac{1}{(1+x^2)^n} dx = \frac{\pi i \operatorname{Res}\left[\frac{1}{(1+z^2)^n}, i\right]}{2}$$

$$\text{同理 } (1+z)^d = 1 + dz + \frac{d(d-1)}{2!} z^2 + \dots + \frac{d(d-1)\dots(d-k+1)}{k!} z^k + \dots$$

(|z| < 1) 对非0与非正整数

当 $d = -n$ 时,

$$\begin{aligned} \frac{1}{(1+z)^n} &= 1 - nz + \frac{n(n+1)}{2!} z^2 + \dots + \frac{(-1)^k n(n+1)\dots(n+k-1) z^k}{k!} + \dots \\ &= 1 - nz + \dots + \frac{(-1)^k n(n+1)\dots(n+k-1)}{(n-1)! k!} z^k + \dots \\ &= 1 - nz + \dots + \frac{(-1)^k C_{n+k-1}^{n-1}}{k!} z^k + \dots \end{aligned}$$

$$\begin{aligned}
 \text{由于 } \frac{1}{(1+z^2)^n} &= \frac{1}{(z+i)^n(z-i)^n} = \frac{1}{(z-i+z+i)^n} \cdot \frac{1}{(z-i)^n} \\
 &= \frac{1}{(z-i)^n} \cdot \frac{1}{(1+\frac{z-i}{z-i})^n} \cdot \frac{1}{(z-i)^n} \\
 &= \frac{1}{2^n i^n} \cdot \frac{1}{(z-i)^n} \cdot [1 + (-n)(\frac{z-i}{z-i}) + \dots + (-1)^{n-1} C_{2n-2}^{n-1} (\frac{z-i}{z-i})^{n-1} + \dots]
 \end{aligned}$$

$$\begin{aligned}
 \frac{1}{z-i} \text{ 的系数是 } C_{-1} &= \frac{1}{2^n i^n} \cdot (-1)^{n-1} \cdot C_{2n-2}^{n-1} \cdot \frac{1}{(z-i)^{n-1}} \\
 &= \frac{(-1)^{n-1}}{2^{2n-1} i^{2n-1}} C_{2n-2}^{n-1} \\
 &= \frac{(-i) C_{2n-2}^{n-1}}{2^{2n-1}}
 \end{aligned}$$

$$\text{故 } J_n = \pi i \cdot C_{-1} = \frac{C_{2n-2}^{n-1} \pi}{2^{2n-1}} = \frac{\pi}{2^{2n-1}} \frac{(2n-2)!}{[(n-1)!]^2}$$

$$n=1 \text{ 时, } J_1 = \frac{\pi}{2}$$

$$n>1 \text{ 时, } \frac{J_{n+1}}{J_n} = \frac{\pi (2n)!}{2^{2n+1} [(n+1)!]^2} / \frac{\pi (2n-2)!}{2^{2n-1} [(n-1)!]^2} = \frac{2n-1}{2n} < 1$$

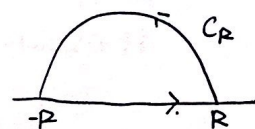
(得到递推关系)

$$\text{例 17. 求 } J_{r,n} = \int_0^{+\infty} \frac{dx}{(r^2+x^2)^n}, r>0, n \in \mathbb{Z}^+$$

$$J_{r,n} = \frac{1}{r^{2n-1}} J_n = \frac{\pi C_{2n-2}^{n-1}}{r^{2n-1} \cdot 2^{2n-1}} = \frac{C_{2n-2}^{n-1} \pi}{(2r)^{2n-1}}$$

$$\text{例 18. 求 } I = \int_{-\infty}^{+\infty} \frac{\cos kx}{x^2+ax+b} dx, a^2>4b, a,b \in \mathbb{R}, k>0.$$

$$I = \int_{-\infty}^{+\infty} \frac{\cos kx}{(x+\frac{a}{2})^2 + \frac{4b-a^2}{4}} dx, \text{ 记 } r^2 = \frac{4b-a^2}{4} > 0$$



$$I = \int_{-\infty}^{+\infty} \frac{\cos kx}{(x+\frac{a}{2})^2 + r^2} dx, \text{ 令 } t = x + \frac{a}{2}, \cos kx = \cos k(t - \frac{a}{2}) = \cos kt \cos \frac{a}{2} + \sin kt \sin \frac{a}{2}$$

$$I = \int_{-\infty}^{+\infty} \frac{\cos kt}{t^2+r^2} dt \cdot \cos \frac{a}{2} + \int_{-\infty}^{+\infty} \frac{\sin kt}{t^2+r^2} dt \cdot \sin \frac{a}{2}$$

$$\text{其中 } \int_{-\infty}^{+\infty} \frac{\sin kt}{t^2+r^2} dt = 0$$

$$\text{故 } I = \cos \frac{a}{2} \int_{-\infty}^{+\infty} \frac{\cos kt}{t^2+r^2} dt = \cos \frac{a}{2} \int_{-\infty}^{+\infty} \frac{e^{ikt}}{t^2+r^2} dt$$

$$\text{因为 } e^{ikt} = \cos kt + i \sin kt, \int_{-\infty}^{+\infty} \frac{i \sin kt}{t^2+r^2} dt = 0$$

$$\Rightarrow I = \cos \frac{a}{2} \times 2\pi i \times \frac{e^{ikt}}{2t} \Big|_{t=ir}$$

$$I = 2\pi i \cos \frac{a}{2} \cdot \frac{e^{-kr}}{2ir} = \frac{\pi \cos \frac{a}{2}}{r e^{kr}}$$

$$\text{由于 } r^2 = \frac{4b-a^2}{4}, I = \frac{\pi \cos \frac{a}{2}}{\frac{\sqrt{4b-a^2}}{2} \cdot e^{\frac{k\sqrt{4b-a^2}}{2}}} = \frac{2\pi \cos \frac{a}{2}}{\sqrt{4b-a^2}} e^{\frac{k\sqrt{4b-a^2}}{2}}$$

一些视频中宣称会考的积分:

$$\oint_{|z|=r>1} \frac{dz}{1+z^n}, \quad \oint_{|z|=r>1} \frac{z^{2n}}{1+z^n} dz, \quad \oint_{|z|=1} \frac{\sin z^n}{z^m},$$

以及标△的题.

一些变形.(例如 $\sin? \cos?$ 等?).

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例 19. $I_{m,n} = \oint_{|z|=1} \frac{1}{(z-a)^n(z-b)^n} dz, |a| \neq 1, |b| \neq 1, a \in \mathbb{C}, b \in \mathbb{C}.$

1. $a=b$ 时, $I_{m,n} = \oint_{|z|=1} \frac{1}{(z-a)^{m+n}} dz = 0.$

2. $a \neq b$ 时,

当 $|a| < 1, |b| < 1$. 令 $t = \frac{1}{z} \Rightarrow z = \frac{1}{t} \Rightarrow dz = (-\frac{1}{t^2}) dt$

$$\begin{aligned} I_{m,n} &= \oint_{|t|=1} \frac{1}{(\frac{1}{t}-a)^n(\frac{1}{t}-b)^n} (-\frac{1}{t^2}) dt \\ &= \oint_{|t|=1} \frac{t^{m+n-2}}{(1-at)^n(1-bt)^n} dt. \end{aligned}$$

在 $|t| \leq 1$ 内, $\frac{t^{m+n-2}}{(1-at)^n(1-bt)^n}$ 处处可导,

故 $I_{m,n} = 0.$

当 $|a| > 1, |b| > 1$. 由 C-G 定理, $I_{m,n} = 0.$

不妨设 $|a| < 1 < |b|$.

$$\begin{aligned} I_{m,n} &= \oint_{|z|=1} \frac{1}{(z-a)^n(z-a+b-b)^n} dz = \oint_{|z|=1} \frac{1}{(z-a)^n} \left[\sum_{k=0}^{+\infty} \frac{(-1)^k C_{m+n-1}^{m-1}}{(a-b)^{m+n}} (z-a)^k \right] dz \\ &= \sum_{k=0}^{+\infty} \frac{(-1)^k C_{m+n-1}^{m-1}}{(a-b)^{m+n}} \oint_{|z|=1} \frac{dz}{(z-a)^{n-k}} \\ &= 2\pi i \frac{(-1)^{n-1} C_{m+n-2}^{m-1}}{(a-b)^{m+n-1}} \end{aligned}$$

综上, $I_{m,n} = \begin{cases} 0 & a=b \\ 0 & |a| < 1, |b| < 1 \\ 0 & |a| > 1, |b| > 1 \\ 2\pi i \frac{(-1)^{n-1} C_{m+n-2}^{m-1}}{(a-b)^{m+n-1}} & |a| < 1 < |b| \end{cases}$

若 $|b| < 1 < |a|$. $2\pi i \frac{(-1)^{m-1} C_{m+n-2}^{n-1}}{(b-a)^{m+n-1}}$

例20. 求 $I_1 = \oint_{|z|=1} \frac{1}{(z-a)^n(z-b)^n} dz$. $|a| \neq 1, |b| \neq 1$.

等同于例19.